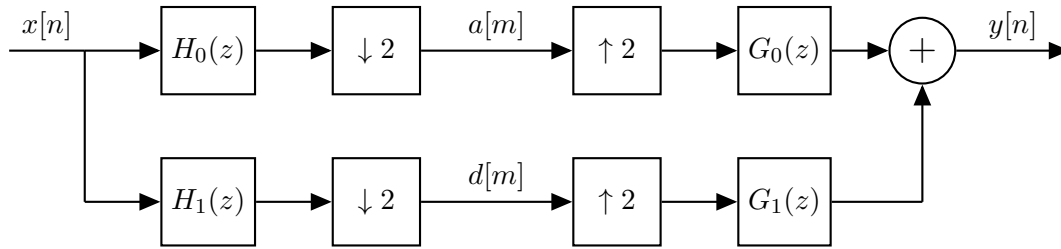


Homework #9

Due: 11:59pm, April 27

Objectives: Study wavelet transform and design of filter banks**1. [Orthogonal PR-FIR]**

We will design orthogonal PR-FIR filters of length $L = 6$ for the following two-channel filterbank system.



- Let $Q(z) = q_2 z^2 + q_1 z + q_0 + q_1^{-1} + q_2 z^{-2}$. Find the coefficients q_0 , q_1 and q_2 so that $A_0(z) = (1 + z^{-1})^{L/2} (1 + z)^{L/2} Q(z)$ satisfies $a_0[2m] = \delta[m]$. Here $a_0[n]$ denotes the inverse z -transform of $A_0(z)$.
- Compute the minimum phase factor $R(z)$ of $Q(z) = R(z)R(z^{-1})$ by spectral factorization.
- Compute $g_0[n]$ from $R(z)$ by using $G_0(z) = (1 + z^{-1})^{L/2} R(z)$.
- Compute $h_0[n]$, $h_1[n]$, $g_1[n]$ from $g_0[n]$ by using the following identities:

$$G_0(z) = H_1(-z), \quad G_1(z) = -H_0(-z), \quad H_1(z) = -z^{-(L-1)} H_0(-z^{-1}).$$

- Show that $g_0[n]$ and $g_1[n]$ have the same energy.
- Show that

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = C \left(\sum_{m=-\infty}^{\infty} (|a[m]|^2 + |d[m]|^2) \right).$$

Express the constant C by using $g_0[n]$.

2. [Reciprocal Zeros of Generalized Linear-Phase FIR Filters]

Let $h[n]$ be a real-valued FIR filter of even length L , with

$$H(z) = \sum_{n=0}^{L-1} h[n] z^{-n}.$$

Assume that $h[n]$ has *generalized linear phase*. Consider the following two cases.

- Symmetric case:** $h[n] = h[L - 1 - n]$.

(b) **Antisymmetric case:** $h[n] = -h[L - 1 - n]$.

Show that in each case, if $z_0 \in \mathbb{C}$ is a zero of $H(z)$, then z_0^{-1} is also a zero of $H(z)$.

3. **[Incompatibility of Orthogonality and Generalized Linear Phase]**

Let $h[n]$ be a real-valued FIR low-pass filter of even length $L \geq 4$, with

$$H(z) = \sum_{n=0}^{L-1} h[n]z^{-n}.$$

Assume that $H(z)$ is the analysis low-pass filter of an *orthogonal wavelet filter bank*, so that the quadrature mirror filter (QMF) condition

$$H(z)H(z^{-1}) + H(-z)H(-z^{-1}) = 2 \quad (1)$$

holds for all $z \neq 0$. Suppose further that $H(z)$ is constructed according to the Daubechies design and admits the factorization

$$H(z) = (1 + z^{-1})^{L/2} R(z),$$

where $R(z)$ is a polynomial in z^{-1} of degree $(L - 2)/2$. By the spectral factorization in the Daubechies construction, $R(z)$ is *minimum phase*, i.e., all its zeros lie strictly inside the unit circle. Show that $H(z)$ *cannot have generalized linear phase*.

You may use the following lemma.

Lemma 1 *If $L \geq 4$ and $H(z)$ satisfies the QMF condition, then $R(z)$ cannot be a power of $(1 + z^{-1})$. In particular, $H(z)$ has at least one zero $z_0 \notin \{\pm 1\}$.*

Hint. Deduce the conclusion by following the steps below.

- Since $\deg R \geq 1$ for $L \geq 4$, Lemma 1 implies that $H(z)$ has at least one zero z_0 satisfying

$$z_0 \notin \{\pm 1\}.$$

- Define

$$P(z) = H(z)H(z^{-1}).$$

Show that $P(z_0) = 0$ implies $P(z_0^{-1}) = 0$.

- Because $z_0 \notin \{\pm 1\}$, exactly one of $\{z_0, z_0^{-1}\}$ lies strictly inside the unit circle, while the other lies strictly outside. Hence, the minimum-phase spectral factorization assigns only one element of $\{z_0, z_0^{-1}\}$ to the zeros of $R(z)$, while the other is assigned to the zeros of $R(z^{-1})$.
- Show that this asymmetric zero assignment contradicts the zero-symmetry required by the generalized linear-phase condition established in Problem 2.

Proof of Lemma 1 Since $L \geq 4$, we have $\deg R = (L - 2)/2 \geq 1$, so $R(z)$ has at least one zero z_0 . If $R(z)$ were a power of $(1 + z^{-1})$, then all zeros of $H(z)$ would be located at $z = -1$. This would imply

$$H(z) = (1 + z^{-1})^{L-1}.$$

Evaluating the QMF condition at $z = 1$, we obtain

$$H(1)^2 + H(-1)^2 = 2.$$

However,

$$H(1) = 2^{L-1}, \quad H(-1) = 0,$$

so

$$H(1)^2 + H(-1)^2 = 2^{2L-2} \neq 2 \quad \text{for } L \geq 4.$$

This contradicts the QMF condition. Hence $R(z)$ cannot be a power of $(1 + z^{-1})$. $\square$